

Name : Ragini Singh

Cohort : JH

Roll No : 150096724023

Subject : Physics Assignment 1

Page No. :
Date : 19/08/26

Physics Assignment-1

Name — Ragini Singh
Cohort — JH
Roll No — 150096724023
Date — 19/08/26

Q.1 A $12\ \Omega$ resistor is connected to a $24\ \text{V}$ supply. Find the Current flowing Through the resistor.

Solution: Given that

$$R = 12\ \Omega$$
$$V = 24$$

we know that — $V = IR$

$$\Rightarrow I = \frac{V}{R} = \frac{24}{12} = 2\ \text{A}$$

Ans.

Q.2 R_1 R_2
 $R_1 + R_2$

Class explanation

Ans. $R_1 + R_2$

Q.3

The diagram shows a rectangular circuit loop. On the left vertical wire is a battery symbol with a '+' sign at the top and a '-' sign at the bottom, labeled '5V'. On the top horizontal wire is a resistor symbol (zigzag line) labeled '20'. On the right vertical wire is another resistor symbol labeled '24'. On the bottom horizontal wire, an arrow points from left to right, labeled 'I'.

Solution: Find the value of I

$$V = IR$$

$$I = \frac{V}{R}$$

$$I = \frac{V}{R_1 + R_2}$$

$$= \frac{5}{2+2} = \frac{5}{4} = 1.25 = 1.25 \text{ A}$$

Q.3. Explain overclocking?? Ans. Explain in brief overclocking.

ans. overclocking means forcing your computer components to run faster than the speed limits set by the manufacturer. By increasing the operating frequency, your hardware processes data quicker. This translates directly into higher frame rates (FPS) and smoother gameplay without buying new parts.

Q.4. A Resistor of 6Ω is connected to $12V$ battery. a charge particle of mass $m = 2\text{kg}$ experience a force equal to the weight following throwing to the resistor. Find the q value?

Solution: Formula = $F = ma$

$$V = IR$$

$$\Rightarrow 12 = I \cdot 6$$

$$\Rightarrow \frac{12}{6} = I$$

$$\Rightarrow 2 = I$$

$$\Rightarrow 2 = 2a$$

$$\Rightarrow a = \frac{2}{2}$$

$$\Rightarrow a = 1 \text{ m/s}^2$$

Q.5. Explain in brief electromagnetism?

Ans. Electromagnetism is a relationship between electricity and magnetism.

example: Voltage regulator modules (VRMs) use inductors to stabilize CPU power.

Q.6. M.C.Q. Type Ques

Q. which of the following correctly describe the magnetic field near along straight wire?

① The field consist of straight lines perpendicular to the wire.

② The field consist of parallel to the wire

③ The field consist of radial lines originating from the wire.

Date: 19/8/26

⑦ The field of Consequent circle Centre on the wire.

Q.2 At the time of short circuit the current in the circuit, vary continuously

⑪ Does not change

(iv) Vary continuously.

(iv) Vary continuously.